13	**(a)**	**(i)**	[Area =] $2xl + \pi x^2$ $l = \pi(r - x)$ so Area = $2x\pi(r-x) + \pi x^2 = 2\pi rx - \pi x^2 = \pi x(2r - x)$	**M1** **A1** **[2]**	**1.1a** **2.4**	Convincing completion (answer given)	Use of $l = \pi(r - x)$ to get formula in terms of x
13	**(a)**	**(ii)**	$ry + hx = hr$ so $x = \dfrac{hr - ry}{h}$ and $\pi x(2r - x) = \pi\left(\dfrac{hr - ry}{h}\right)\left(\dfrac{2rh - hr + ry}{h}\right)$ $\pi\left(\dfrac{hr - ry}{h}\right)\left(\dfrac{hr + ry}{h}\right) = \dfrac{\pi}{h^2}\left(h^2r^2 - r^2y^2\right)$ $= \dfrac{\pi r^2}{h^2}\left(h^2 - y^2\right)$	**M1** **A1** **[2]**	**3.1a** **2.2a**	Finding x in terms of y correctly and using it Convincing completion (answer given)	
13	**(b)**	**(i)**	$y = 0$ so area = $\dfrac{\pi r^2}{h^2}\left(h^2\right) = \pi r^2$	**B1** **[1]**	**3.5a**		
13	**(b)**	**(ii)**	$y = h$ so area = $\dfrac{\pi r^2}{h^2}\left(h^2 - h^2\right) = 0$	**B1** **[1]**	**3.5a**		

							Condone